

Research Flowchart

Gravity Model and Network Analysis in International Trade (1995-2022)



STEP 1 Data Collection and Preparation

OECD ICIO

- Input-output tables
- Inter-sectoral linkages
- 1995-2020 period

CEPII BACI

- Bilateral trade data
- Product & product group level
- 1995-2022 period



STEP 2 Multiplex Network Construction

OECD ICIO Network

- Layers: sectors
- Inter-layer links: ✓
- Cross-sectoral flows

CEPII BACI Network

- Layers: products/groups
- Inter-layer links: ✗
- Dynamic structure



STEP 3 Network Metrics Calculation

Network-level metrics

Node-level metrics

Annual breakdown

Product/sector level



STEP 4 Gravity Model Estimation



GDP Data
Origin & destination



Distance
Between countries



Coefficients
 β parameters



STEP 5 Clustering and Forecasting



Distance coefficient clustering

Based on time series characteristics



Forecasting to 2030

Group-level projections



Analysis: Which products/sectors belong to each cluster?



STEP 6 Correlation and Causality Analysis

Y Dependent Variables

- Distance effect (β_{distance})
- Origin GDP ($\beta_{\text{GDP,origin}}$)
- Destination GDP ($\beta_{\text{GDP,dest}}$)

X Independent Variables

- Network structural parameters
- Node centralities
- Multiplex network metrics